ROBUST NEURAL DECODING UNDER MISSING CHANNELS

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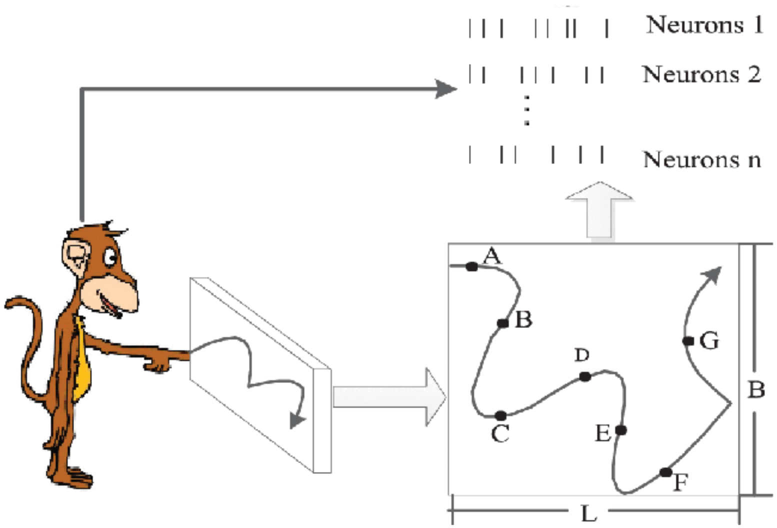

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LONG ISLAND UNIVERSITY

OUTLINES

1. Problem Overview
2. Dataset Overview
3. Problem Formulation
4. Methodology
5. Key Experiments
6. Results
7. Key Insights
8. Conclusion & Future work

PROBLEM OVERVIEW

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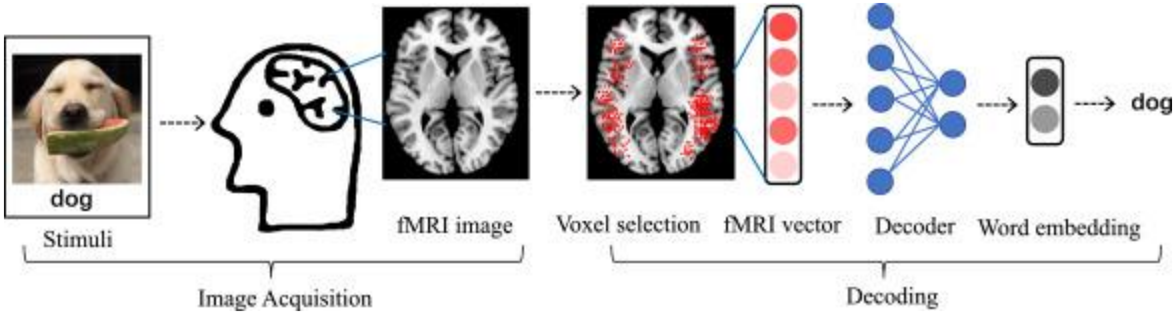
- **Neural decoding = brain → movement**

- **Applications:**
  - ➢ Prosthetics
  - ➢ Assistive tech

- **Challenge:**
  - ➢ Noisy signals
  - ➢ Missing channels



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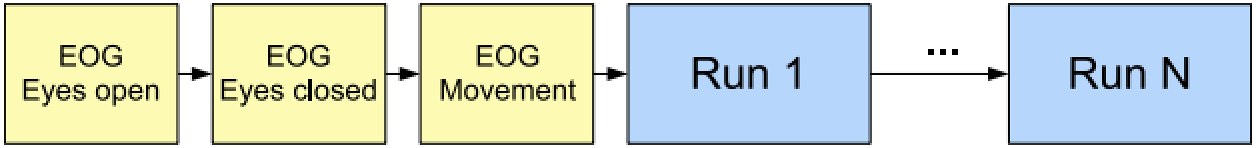
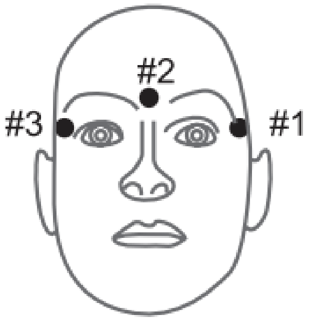
DATASET OVERVIEW

- **Dataset**: BCI Competition IV Dataset 4
- 62 channels
- **Output**: 5 finger positions
- **Subjects**: S1, S2, S3
- Time-series data

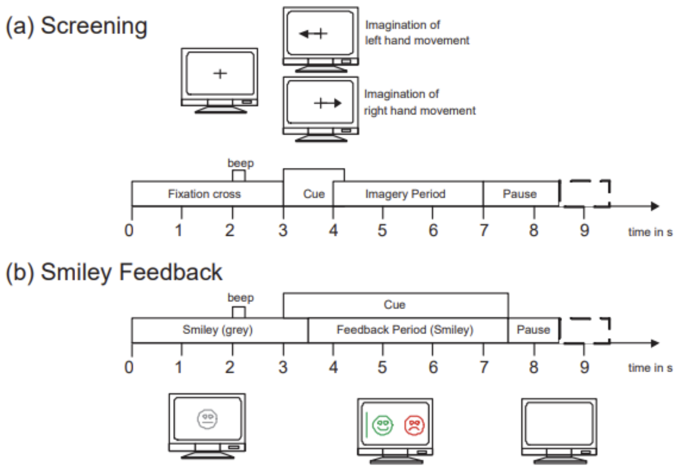




R. Leeb et al. BCI Competetion 2008

PROBLEM FORMULATION

- **Input**: neural signals (62 channels)
- **Output**: finger positions (5 values)
- **Type**: Regression problem
- **Metric**: R^2 score



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METHODOLOGY

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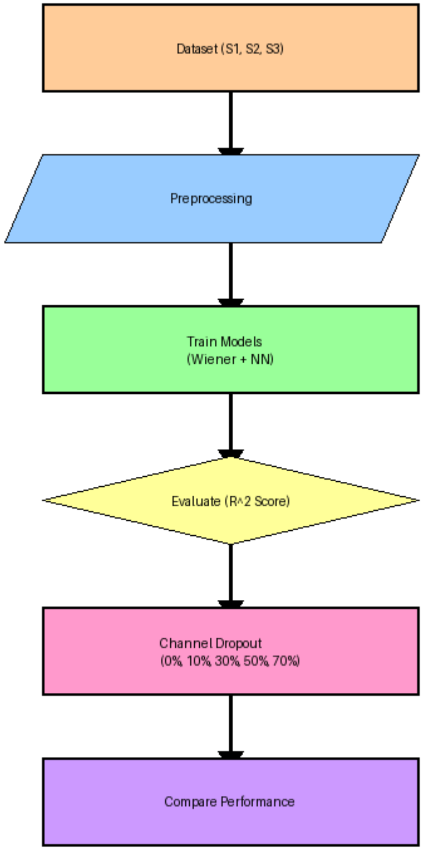
- **Models:**
  - ✓ Wiener Filter (Linear)
  - ✓ Neural Network (Nonlinear)
- **Preprocessing:**
  - ✓ StandardScaler
- **Experiment:**
  - ✓ Channel Dropout (0% → 70%)



KEY EXPERIMENT

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- **Channel dropout levels:**
  - 0%, 10%, 30%, 50%, 70%

- **Goal:**
  - Test robustness

- **Compare:**
  - Wiener vs Neural Network

RESULTS

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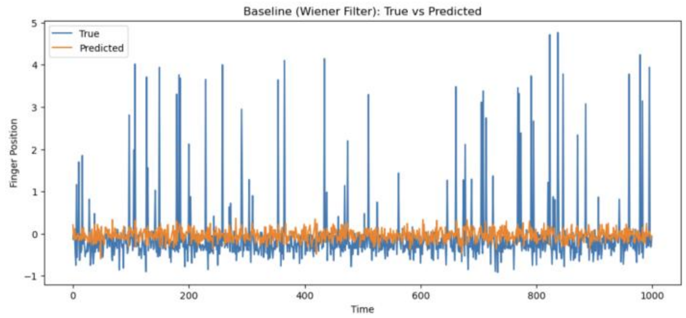
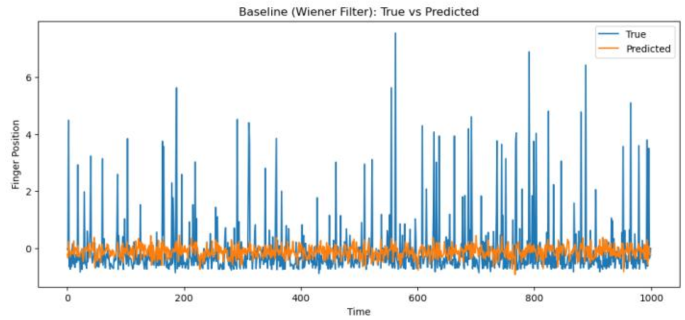
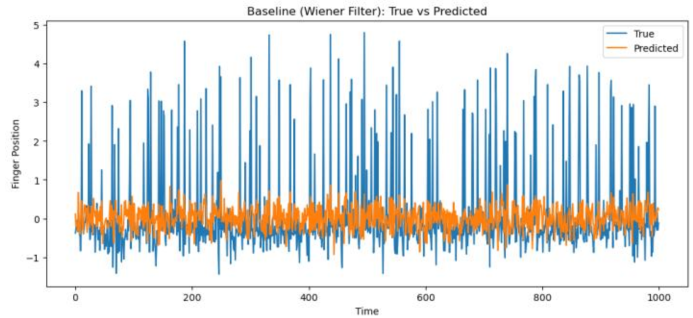
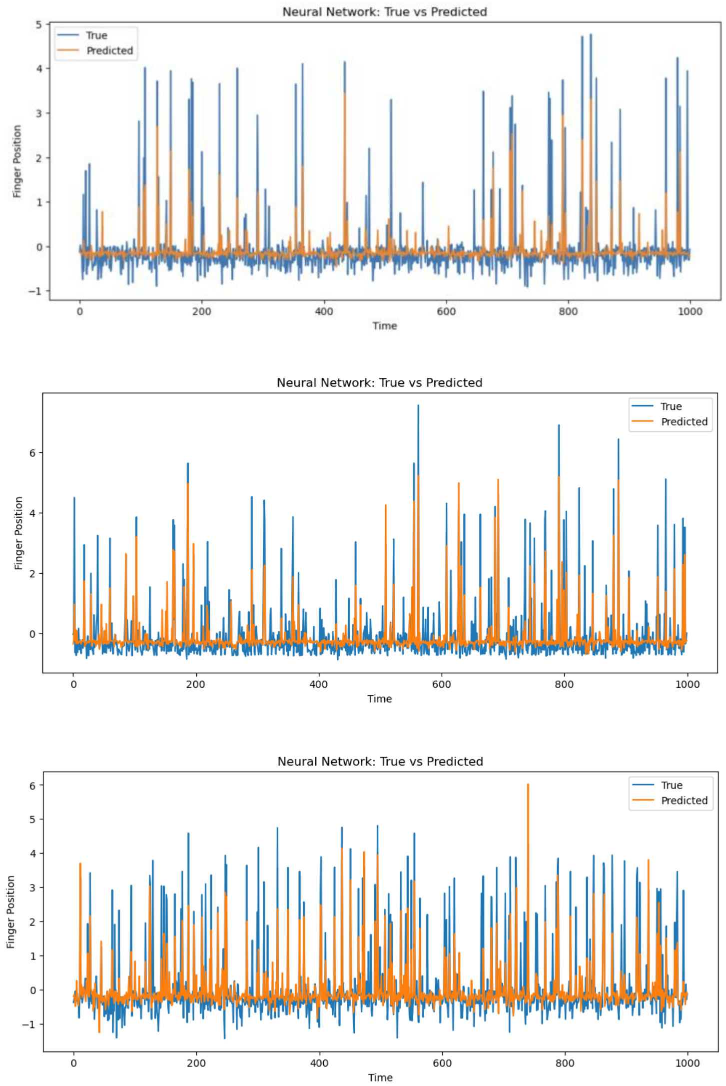






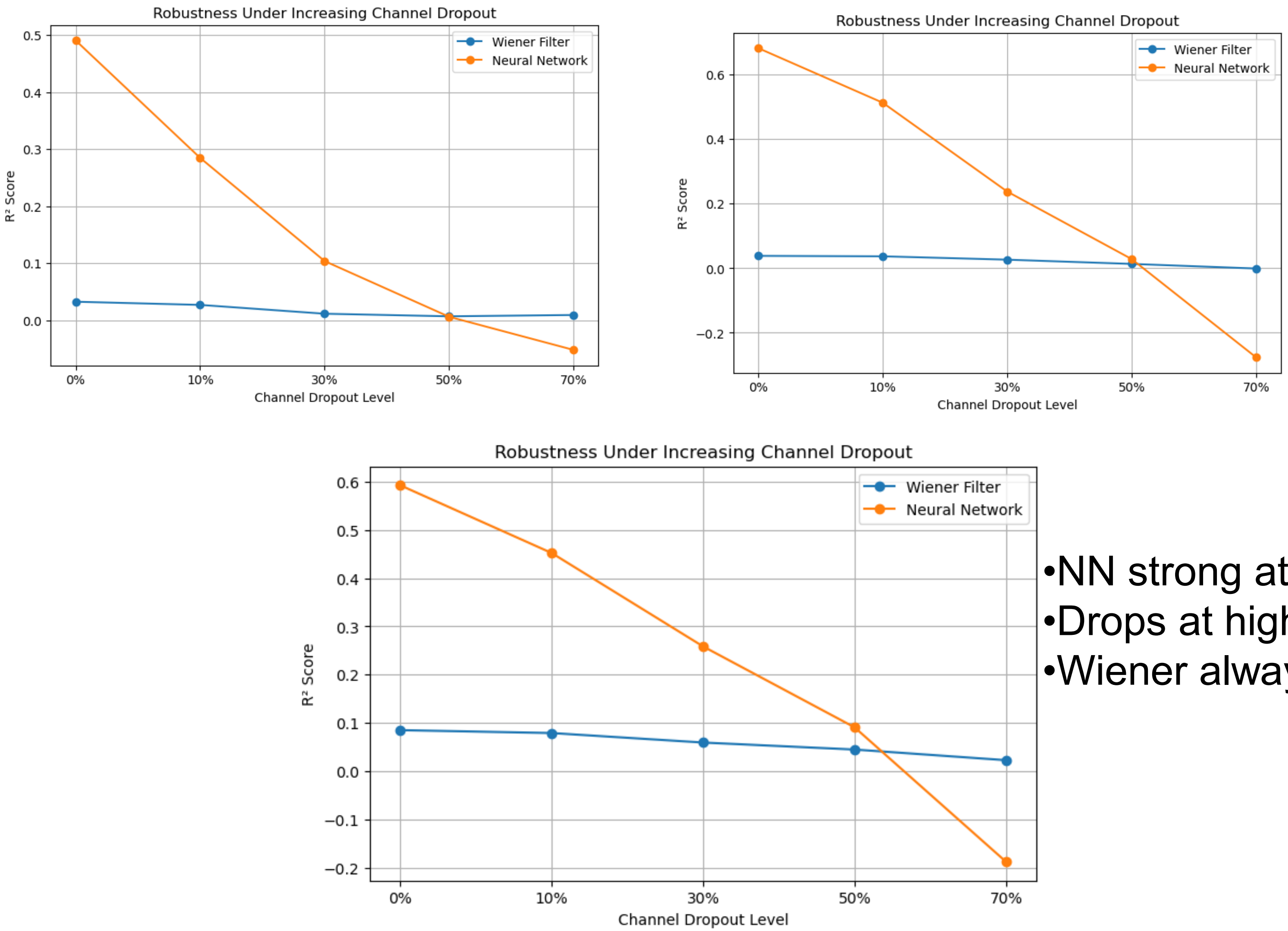
Wiener Filter True Vs Predicted graph (for S1, S2 and S3)

RESULTS



Neural Network True Vs Predicted graph (for S1, S2 and S3)

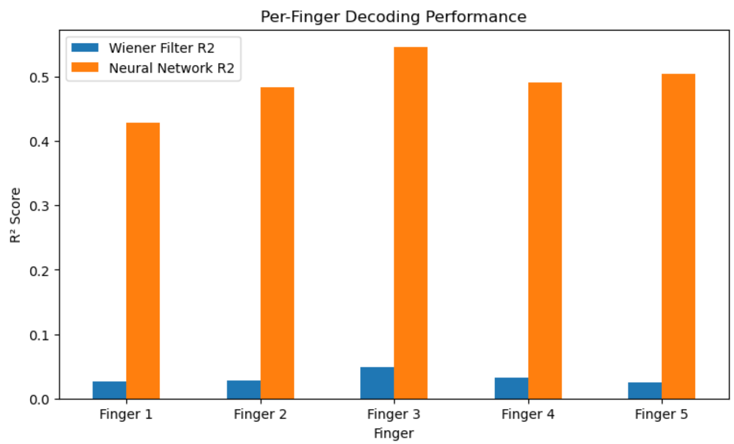
RESULTS

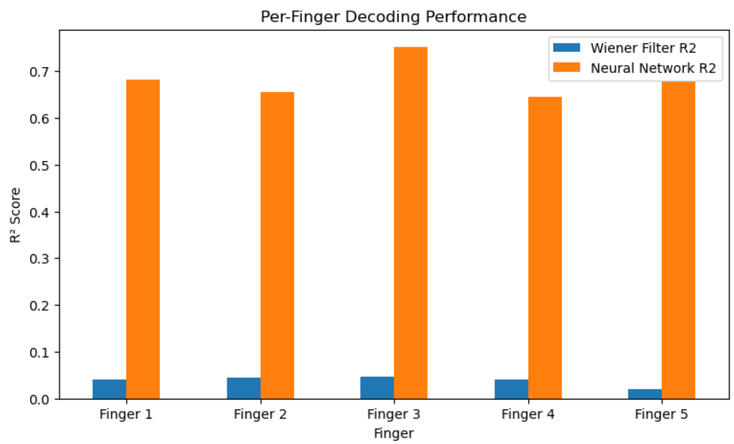


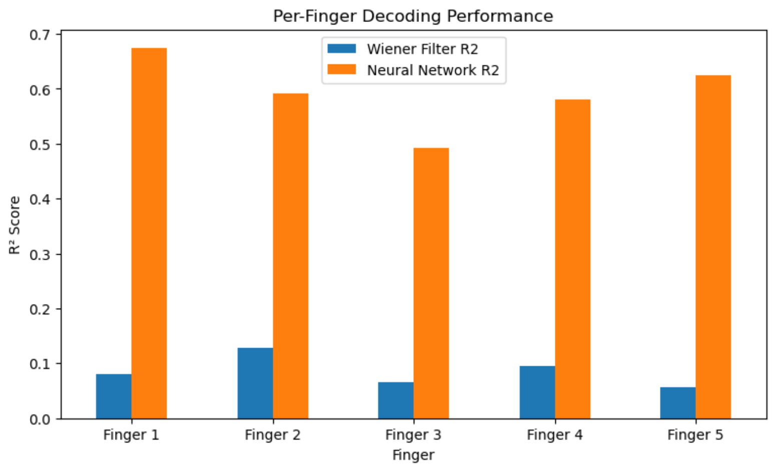
- NN strong at low dropout
- Drops at high dropout
- Wiener always low

Robustness under increasing channel dropout (for S1, S2 and S3)

RESULTS







- NN better for all fingers
- consistent advantage

Plot of Per Finger Decoding Performance (for S1, S2 and S3)

RESULTS

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Dropout Level	S1 Wiener R^2	S1 NN R^2	S2 Wiener R^2	S2 NN R^2	S3 Wiener R^2	S3 NN R^2
0%	0.032	0.491	0.038	0.682	0.085	0.593
10%	0.027	0.285	0.037	0.513	0.079	0.452
30%	0.011	0.104	0.027	0.238	0.059	0.258
50%	0.007	0.006	0.013	0.029	0.045	0.091
70%	0.009	-0.052	-0.001	-0.276	0.023	-0.188

- NN >> Wiener
- Consistent across S1, S2, S3

Finger	S1 (R^2)		S2 (R^2)		S3 (R^2)	
	Wiener	NN	Wiener	NN	Wiener	NN
F1	0.027	0.429	0.041	0.682	0.080	0.674
F2	0.028	0.483	0.045	0.656	0.128	0.591
F3	0.048	0.546	0.046	0.752	0.066	0.493
F4	0.033	0.491	0.040	0.645	0.095	0.580
F5	0.026	0.504	0.020	0.678	0.056	0.625

KEY INSIGHTS

- NN performs best
- Performance ↓ with dropout
- **Challenges:**
  - High dimensionality
  - Nonlinear mapping
  - Missing channels
- **Robustness strategies:**
  - Nonlinear models
  - Dropout training

CONCLUSION & FUTURE WORK

- NN > Wiener
- Robust under moderate dropout

- **Limitations:**
  - High dropout failure

- **Future Work:**
  - RNN implementation
  - Data augmentation
  - Better robustness methods

THANK YOU!!

Questions